PRACTICAL 2

Email Spam Detection using Binary Classification, KNN and SVM

Top 60 Viva Questions & Answers | Important concepts, tricky questions, formulas and last-minute recall

Class labels: *0 = Normal / Not Spam | 1 = Abnormal / Spam*

Use this as the revision sheet after learning the practical. Code syntax is intentionally minimized; focus on concepts, reasoning and interpretation.

A. Machine Learning and Classification

1. What is Machine Learning?

Answer: A branch of AI that enables computers to learn patterns from data and make predictions without being explicitly programmed for every case.

2. What is supervised learning?

Answer: A type of machine learning in which a model learns from labelled examples containing inputs and their correct outputs.

3. What is classification?

Answer: A supervised learning task that assigns data to predefined categories or classes.

4. What is binary classification?

Answer: Classification with exactly two possible classes. Here, 0 represents Not Spam and 1 represents Spam.

5. Why is spam detection a classification problem?

Answer: Because the required output is one of two categories: spam or not spam.

6. What is the difference between classification and regression?

Answer: Classification predicts categories; regression predicts continuous numerical values, such as prices or temperatures.

7. What are features and labels?

Answer: Features (X) are the input variables used for prediction. Labels (y) are the correct target outputs the model learns to predict.

8. What are training data and testing data?

Answer: Training data teaches the model. Testing data evaluates it on unseen examples. This practical uses an 80:20 split.

B. Dataset and Text Preprocessing

9. Why is preprocessing necessary?

Answer: It handles missing, inconsistent or unsuitable data and converts it into a form that machine learning algorithms can use.

10. Why must raw text be converted into numbers?

Answer: Machine learning algorithms perform mathematical operations on numerical representations; they cannot directly use ordinary text as model features.

11. What is TF-IDF?

Answer: Term Frequency–Inverse Document Frequency. It converts text into numerical feature vectors by assigning importance weights to words.

12. What is Term Frequency (TF)?

Answer: It measures how frequently a word appears in a particular document, often normalized by the document's total word count.

13. What is Inverse Document Frequency (IDF)?

Answer: It measures how uncommon a word is across the document collection. Words appearing in many documents generally receive lower IDF weights.

C. Train-Test Split, Scaling and Data Leakage

14. Why do we split the dataset?

Answer: To train on one portion and evaluate on separate, unseen data so that performance better reflects generalization.

15. Why is test_size set to 0.2?

Answer: It reserves 20% of the data for testing and leaves 80% for training.

16. What is stratify=y and why is it used?

Answer: It preserves approximately the same class proportions in training and testing sets, which is useful when classes are imbalanced.

17. What is data leakage?

Answer: Data leakage occurs when information from the test set influences training or preprocessing, causing an overly optimistic evaluation.

18. Why is TF-IDF fitted only on training text?

Answer: The vocabulary and IDF weights must be learned from training data only. Test text is transformed using those learned values to prevent leakage.

19. What is the difference between fit_transform() and transform()?

Answer: fit_transform() learns a transformation from data and applies it. transform() applies an already learned transformation to new data.

20. What is feature scaling?

Answer: It puts numerical features on comparable scales so features with large numeric ranges do not dominate distance calculations or optimization.

21. What is StandardScaler?

Answer: It standardizes a feature by subtracting its training mean and dividing by its training standard deviation, producing values centered around zero with unit variance.

D. K-Nearest Neighbors (KNN)

22. What is KNN?

Answer: K-Nearest Neighbors is a supervised algorithm that predicts a new sample's class using the classes of its K closest training samples.

23. How does KNN classify an email?

Answer: It represents the email as a feature vector, measures its distance from training emails, finds the K nearest examples and predicts the majority class.

24. What does K=5 mean in this practical?

Answer: The classifier considers the five nearest training samples when predicting the class of a test email.

25. How does KNN decide the final class?

Answer: It uses majority voting among the K nearest neighbours. The class with the most votes is predicted.

26. What is Euclidean distance?

Answer: A distance measure that calculates straight-line distance between two feature vectors in the feature space.

27. Write the Euclidean distance formula and explain it.

Answer: d(x,y) = sqrt(sum from i=1 to n of (x_i - y_i)^2). Here x_i and y_i are corresponding feature values and n is the number of features.

28. What happens when K is too small?

Answer: The model becomes sensitive to noise and individual training examples, which can cause overfitting.

29. What happens when K is too large?

Answer: The decision becomes overly influenced by many distant samples and may smooth away local patterns, causing underfitting.

30. Why is KNN called a lazy learner?

Answer: It does little model-building during training and mainly stores training examples; much of the computation happens when making predictions.

31. What are the advantages and limitations of KNN?

Answer: It is simple and intuitive, but prediction can be slow on large datasets, it is sensitive to scaling and distance choice, and high-dimensional text data can make distance less informative.

E. Support Vector Machine (SVM)

32. What is SVM?

Answer: Support Vector Machine is a supervised algorithm that finds a decision boundary separating classes, aiming to maximize the margin between them.

33. What is a hyperplane?

Answer: A decision boundary that separates classes. It is a line in two dimensions, a plane in three dimensions, and a hyperplane in higher dimensions.

34. What is a margin in SVM?

Answer: The distance between the decision boundary and the closest training points from either class. SVM aims to maximize this margin.

35. What are support vectors?

Answer: The training points closest to the decision boundary. They are the critical points that determine the position of the separating boundary.

36. What is a kernel in SVM?

Answer: A kernel function allows SVM to model different decision boundaries, including nonlinear boundaries, by computing relationships in a transformed feature space.

37. Name common SVM kernels.

Answer: Linear, polynomial, radial basis function (RBF) and sigmoid.

38. Why does this practical use kernel='linear'?

Answer: A linear SVM is often effective for high-dimensional sparse text features such as TF-IDF vectors and is a sensible baseline for text classification.

39. What is the difference between linear and nonlinear SVM?

Answer: A linear SVM uses a straight hyperplane in the original feature space. Nonlinear SVM uses a kernel to create a more flexible decision boundary.

40. What are the advantages and limitations of SVM?

Answer: SVM can work well in high-dimensional spaces and often performs strongly on text data. Parameter selection can be difficult and training may be expensive on very large datasets.

41. What is the difference between KNN and SVM?

Answer: KNN predicts from nearby stored examples and is a lazy learner. SVM learns a decision boundary during training and predicts using that boundary.

F. Model Evaluation and Confusion Matrix

42. What is accuracy?

Answer: The proportion of all predictions that are correct. Accuracy = (TP + TN) / (TP + TN + FP + FN).

43. What is a confusion matrix?

Answer: A table comparing actual and predicted classes. For binary classification it counts true positives, true negatives, false positives and false negatives.

44. What is a True Positive (TP)?

Answer: A spam email correctly predicted as spam.

45. What is a True Negative (TN)?

Answer: A normal email correctly predicted as not spam.

46. What is a False Positive (FP)?

Answer: A normal email incorrectly predicted as spam. An important genuine email may be sent to the spam folder.

47. What is a False Negative (FN)?

Answer: A spam email incorrectly predicted as normal, allowing unwanted or potentially harmful mail into the inbox.

48. What is precision?

Answer: Of all emails predicted as spam, the proportion that are actually spam. Precision = TP / (TP + FP).

49. What is recall?

Answer: Of all actual spam emails, the proportion correctly detected. Recall = TP / (TP + FN).

50. What is F1-score?

Answer: The harmonic mean of precision and recall: F1 = 2 × (Precision × Recall) / (Precision + Recall). It balances both measures.

G. Important Tricky and Applied Concepts

51. Why can accuracy be misleading for spam detection?

Answer: If most emails are normal, a model can predict every email as normal and still get high accuracy while detecting no spam. Precision, recall and F1-score reveal more.

52. What is an imbalanced dataset?

Answer: A dataset in which one class has many more examples than another, such as 950 normal emails and 50 spam emails.

53. Which is more important in spam detection: precision or recall?

Answer: It depends on the goal. High precision avoids putting genuine emails into spam; high recall catches more spam. A practical system balances both based on the cost of each error.

54. What is overfitting?

Answer: When a model learns training examples and noise too closely, performing well on training data but poorly on unseen data.

55. What is underfitting?

Answer: When a model is too simple or insufficiently trained to learn important patterns, resulting in poor performance on both training and test data.

56. Why can KNN struggle with TF-IDF text features?

Answer: TF-IDF vectors can have thousands of dimensions and are sparse. Distances may become less informative in high-dimensional spaces, and prediction can be computationally expensive.

57. Why is linear SVM often a strong text-classification baseline?

Answer: Text data is commonly represented by high-dimensional sparse vectors. Linear SVM can find an effective separating boundary in this space.

58. How can KNN performance be improved?

Answer: Tune K, test suitable distance metrics, scale numerical features where appropriate, and use cross-validation to select settings.

59. How can SVM performance be improved?

Answer: Tune parameters such as C and, for nonlinear kernels, gamma; compare kernels and use cross-validation.

60. How does this practical decide which model is better, and what is its limitation?

Answer: It compares the two models' accuracy and selects the higher one. This is simple, but it may overlook precision, recall and F1-score, especially with imbalanced classes.

Formula Sheet – Must Remember

These are the key formulas used by the text representation, KNN and model evaluation concepts in this practical.

Concept	Formula
TF (basic form)	TF(t,d) = count of term t in document d / total terms in document d
IDF (basic form)	IDF(t) = log(N / DF(t))
TF-IDF	TF-IDF(t,d) = TF(t,d) × IDF(t)
Euclidean distance	d(x,y) = sqrt[Σ(i=1 to n) (x_i – y_i)^2]
Standardization (Z-score)	z = (x – μ) / σ
Accuracy	Accuracy = (TP + TN) / (TP + TN + FP + FN)
Precision	Precision = TP / (TP + FP)
Recall (Sensitivity)	Recall = TP / (TP + FN)
F1-score	F1 = 2 × (Precision × Recall) / (Precision + Recall)
Error rate	Error rate = (FP + FN) / (TP + TN + FP + FN)

Symbols used in the formulas

Symbol	Meaning
N	Total number of documents/emails.
DF(t)	Number of documents containing term t.
TP	Spam correctly classified as spam.
TN	Normal email correctly classified as normal.
FP	Normal email incorrectly classified as spam.
FN	Spam incorrectly classified as normal.
x	Original feature value.
μ	Mean of the feature, calculated from training data.
σ	Standard deviation of the feature, calculated from training data.
n	Number of features or dimensions.
K	Number of nearest neighbours considered by KNN.

Last-Minute Recall

- **Binary labels:** 0 = Not Spam; 1 = Spam.
- **TF-IDF:** turns email text into weighted numerical vectors; TF measures within-document frequency, IDF reduces the weight of words common across documents.

• **Data split:** 80% training, 20% testing; stratify preserves class proportions; fit TF-IDF on training data only to avoid leakage.

• **KNN:** finds K nearest training examples and uses majority voting. This practical uses K = 5.

• **SVM:** finds a maximum-margin decision boundary; support vectors are the closest critical training points. This practical uses a linear kernel.

• **Confusion matrix:** TP = spam correctly caught; TN = normal correctly accepted; FP = genuine email wrongly marked spam; FN = spam missed.

• **Metrics:** accuracy measures overall correctness; precision measures reliability of spam predictions; recall measures how much actual spam is caught; F1 balances precision and recall.

• **Tricky point:** High accuracy can be misleading on imbalanced data. Compare precision, recall and F1 as well.

• **Graph question:** The code does not plot a KNN graph. KNN calculates distances internally between numerical feature vectors in a multidimensional feature space.